Cross-Sectional Alpha Factors in Crypto: 2+ Sharpe Ratio Without Overfitting

MARK ARON SZULYOVSZKY AND DANIEL SZEMEREY

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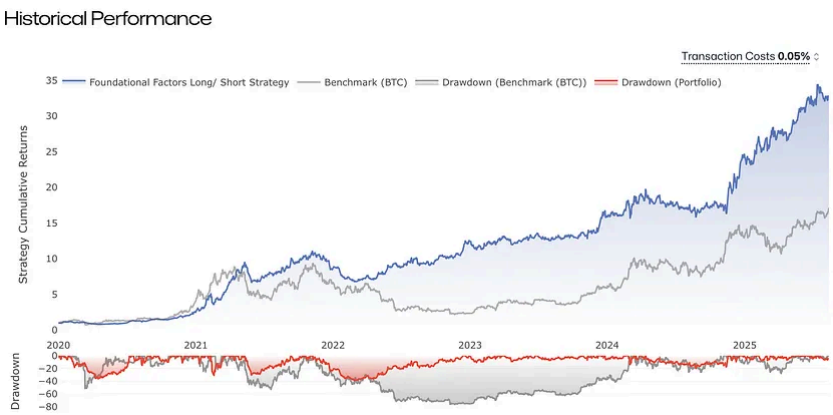

In the early '90s, the quant forefathers (Fama and French) introduced their now-canonical factor models: first three, then five, and eventually seven, explaining muc of the variation in US equity returns. Today, these models are used to understand w easy-to-replicate risk factors managers are being exposed to. Allocators will want t pay for only the truly unique return sources that can't be replicated with a couple o lines of SQL.

Crypto markets are currently in the state where many of these "traditional" factors be still considered alpha, with considerable return potential.

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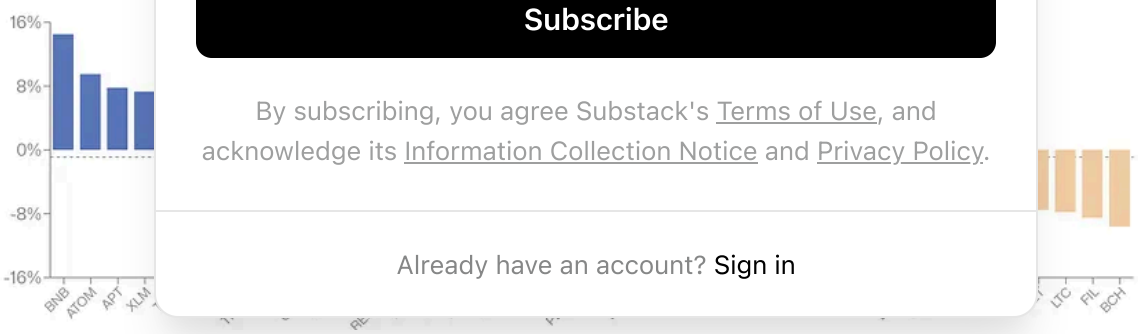
by design, they have 50-50% long and short exposure at any point in time.

How a Long/Short Factor Portfolio Works — in Brief

1. **Define the universe** – e.g., the top 50 digital assets by market cap, updated o rolling basis to
2. **Choose a fact** ...e percentag return over the
3. **Rank the asse**
4. **Go long the le** ...ng position the top 20% a ...%.

This structure isola ...ral.



A typical allocation for a given day (targeting 200% gross exposure)

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Cross-sectional Momentum

Cross-sectional momentum is a well-documented anomaly: assets that outperform over a given formation period ("winners") tend to continue outperforming those tha underperform ("losers")—at least in equities.

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periods) and decay rapidly, sometimes reversing within months.

- **Crash-prone behavior** – Momentum crashes in crypto are typically triggered violent reversals in a handful of assets (e.g., short-squeezed losers), rather tha broad market shifts.
- **Large-cap dominance** – Most of the returns come from top-40 market-cap coins; smaller tokens suffer from liquidity constraints and higher idiosyncratic r
- **Regime sensitivity** – Performance weakens in low-sentiment market regimes.

There's clear evidence that even the simplest **long-only, time-series momentum strategy** on Bitcoin have consistently outperforms in risk-adjusted terms - but it works well in the cross-section as well.

The returns of the long/short cross-sectional, simple momentum portfolio - interactive version.

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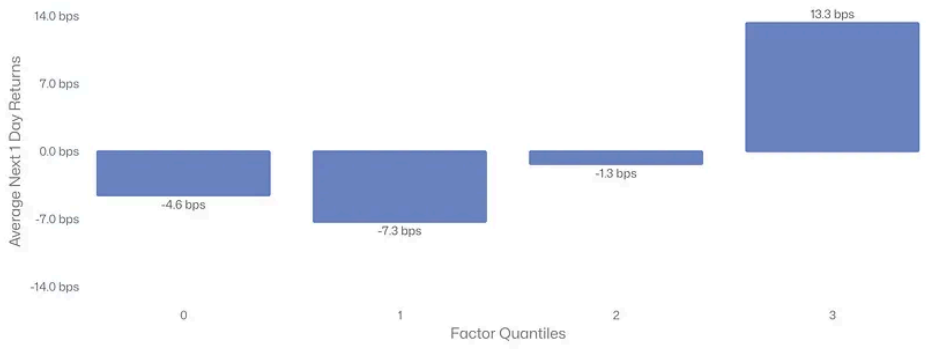

Average next day returns per factor quantile.

Carry in the cross-section

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Unlike momentum, the carry factor traces its roots to commodity markets, where futures often trade at sustained premiums or discounts to spot prices due to factor like storage costs, seasonal patterns, and supply constraints. By expiry, these gaps must close, creating a well-defined arbitrage opportunity.

In crypto, carry is most easily measured via **perpetual futures funding rates**—periodic payments between traders designed to keep perpetual prices anchored to spot. When perpetuals trade above spot, longs pay shorts; when they trade below, shorts pay longs.

There are two ways to implement carry in this context:

- **Hard arbitrage** – Short assets with high funding rates, buy the same notional exposure in spot, and hedge out any price risk. Profits come directly from the

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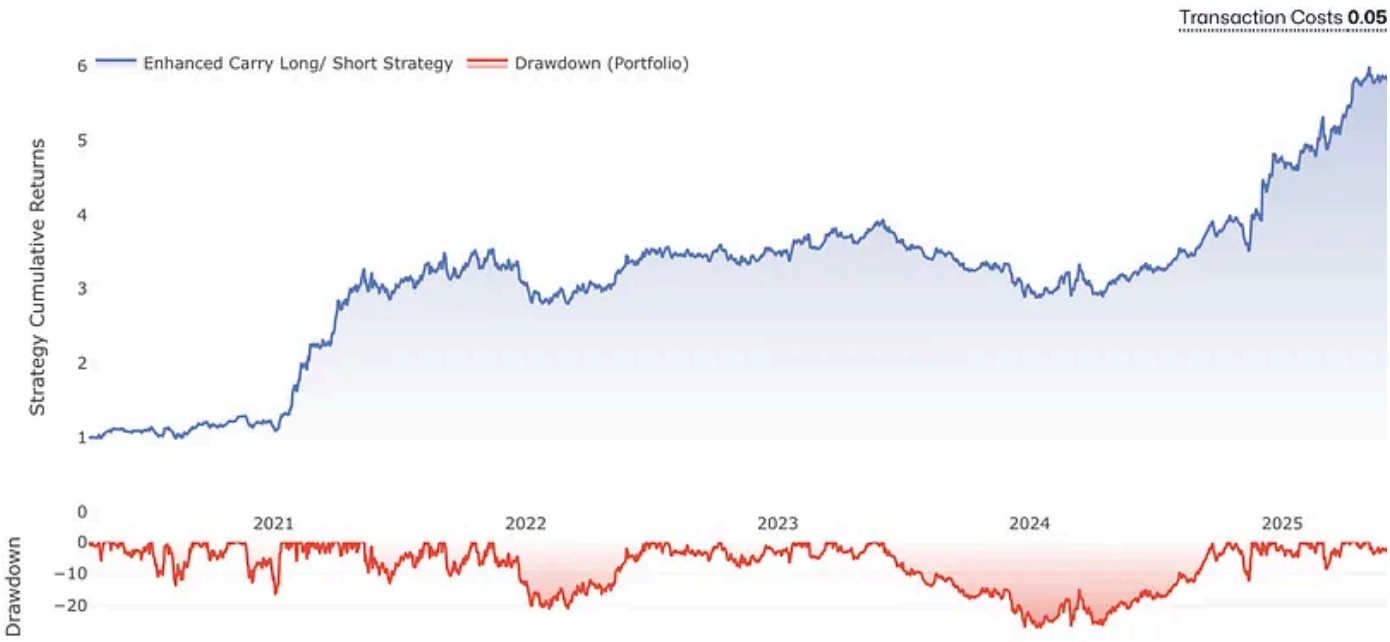
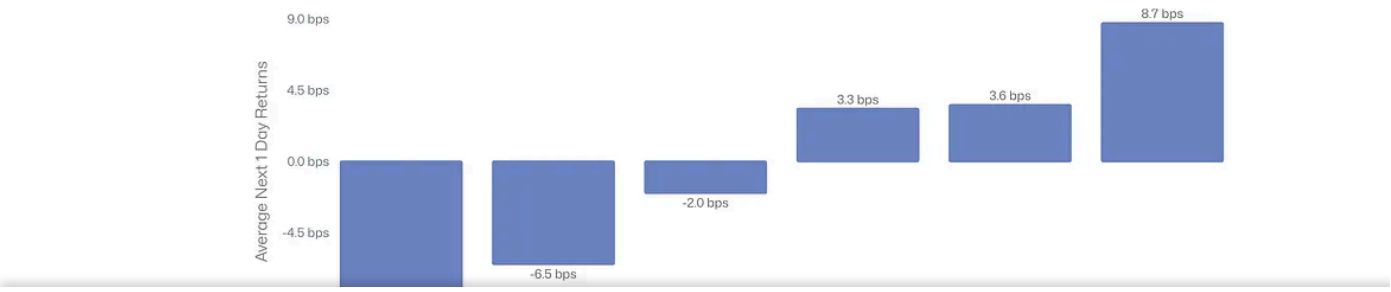
Here, returns come from both the funding spread and differences in expected asset performance.

Funding rates vary across exchanges, so we use an **open-interest-weighted composite** to avoid exchange-specific distortions and better capture the underlyin carry signal.



Returns of the long/short, cross-sectional "enhanced carry" portfolio - interactive version



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The Caveats

Building **crypto-specific, market-neutral, cross-sectional portfolios** is far from straightforward. In equities, short availability is a primary constraint; in crypto, the bigger hurdle is **survivorship bias**—it's often difficult to source reliable data for delisted or inactive tokens.

Beyond that, most crypto assets—outside of a handful of large-cap names—behave more like **penny stocks** than mid-cap companies: thinly traded, highly volatile, and frequently subject to pump-and-dump dynamics. Historical testing before ~2020 is limited use, as the market only reached a comparable level of maturity in the past fi years.

Even in this recent period, creating a stable, tradable universe is challenging. We us market cap ranking with additional **volume, volatility, and availability filters**, with smoothing to limit turnover and avoid overreacting to temporary shifts in liquidity (memecoin spikes).

Another key complication is the **dispersion of volatility** within the tradable univers In equities, volatility differences are significant; in crypto, they're extreme - Bitcoin the average memecoin might as well be different asset classes. Equal notional long and short positions can still leave the portfolio heavily tilted toward the more volatil side. To address this we apply **inverse volatility weighting** at the portfolio level so highly volatile tokens get proportionally smaller allocations.

But what's the alternative to cross-sectional strategies? Potentially overfitting spec signals to individual assets, or abandoning market neutrality entirely.

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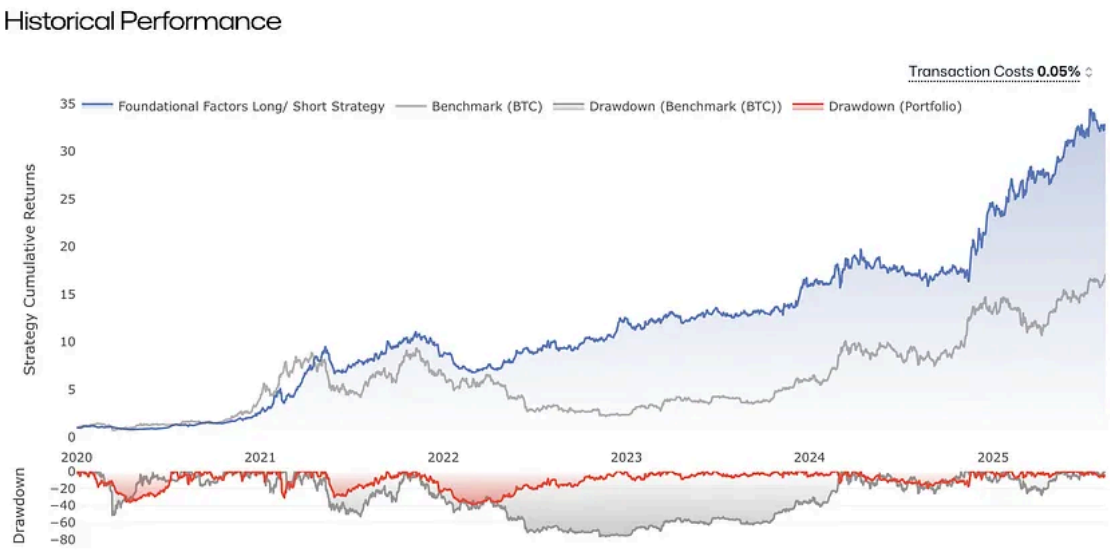
only free lunch" - but diversification works **particularly well** in the context of multi-factor portfolios. The combination doesn't need to be particularly sophisticated eitl even a simple arithmetic average of signals can deliver **tangible benefits**:

- **Reduced turnover** – When factors disagree, offsetting positions naturally can out.
- **Greatly improved risk-adjusted returns** – Uncorrelated return streams smoo the equity curve and raise Sharpe.

When we put these **2** (Momentum + Enhanced Carry) established, well-known fact in a single, portfolio, we get something we call "Foundational" portfolio:



Link to the interactive tearsheet

How Far Can This Scale?

Diversification may be the only free lunch, but even free lunches have limits. Once y

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As always, happy to connect on LinkedIn and talk shop!

* The Sharpe Ratio of "Foundational" is actually less than 2, but we track other mul factor portfolios with considerably higher risk-adjusted returns.

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clipping weights to avoid "over dominance'), instead of simple equal weighted?

LIKE (2) REPLY

1 reply by Mark Aron Szulyovszky

hashmash 17 Aug 2025

You mentioned these are market neutral. I don't see how you are neutralizing them.

Just because you do quantiles or rank-weighted doesnt mean they are market neutral.

E.g. for a basic momentum factor, trading quantiles/rank-weight would make it momentum ne but not necesarily market neutral, since momentum and market returns can (and good chanc correlated.

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